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MILITARY SPECIFICATION

HORNS, SOUND-WARNING, AIR-ACTUATED; HORNS, ELECTRICALLY-ACTUATED; AND BUZZERS, ELECTRICALLY-ACTUATED

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is applicable for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horns and buzzers used in 12- or 24-volt (V) direct current (dc) circuits as warning signals in tactical vehicles, combat vehicles, and other equipment (see 6.1).

1.2 Classification. Horns and buzzers covered by this specification shall be of the following types and classes, as specified (see 6.2 and 6.5):

Type I	Class 1 - Horn, air-actuated, electrically controlled (24 V dc).
	Class 2 - Horn, air-actuated, electrically controlled (12 V dc).
Type II	Class 1 - Horn, electrically-actuated (24 V dc).
	Class 2 - Horn, electrically-actuated (12 V dc).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2590

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Type III Class 1 - Buzzer, electrically-actuated (24 V dc).
Class 2 - Buzzer, electrically-actuated (12 V dc).

Type IV - Horn, electrically-actuated (12 V dc), non-waterproof.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings.

MILITARY

MIL-F-13927	- Fungus Resistance Test, Automotive Components.
MIL-E-17555	- Electronic and Electrical Equipment Accessories and Provisional Item (Repair parts): Packaging of.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-P-53022	- Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.
MIL-C-53039	- Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MS51074	- Horn, Electrical - Automotive Type, 12 or 24 Volts DC.
MS51301	- Horn, Sound-Warning, Air-Actuated, Electrically Controlled, 12 or 24 Volts DC.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2). the issues shall be those cited in the solicitation.

DRAWINGS

ARMY

7728527	- Horn, Commercial.
8675930	- Buzzer.

(Copies of drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish horns or buzzers, of the specified type and class (when applicable) (see 6.2), which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the units to be furnished to the Government. All subsequent horns and buzzers delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

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3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Materials shall be free of defects which adversely affect performance or serviceability of the finished product (see 4.7.1).

3.2.1 Recycled, virgin and reclaimed materials. There are no requirements for the exclusive use of virgin materials. The use of recycled or reclaimed (recovered) materials is acceptable provided that all other requirements of this specification are met (see 6.4.2).

3.3 Design and construction. Horns and buzzers shall be designed and constructed in accordance with the applicable drawings or standards listed in table I (see 4.7.1, 4.7.2 and 6.2).

TABLE I. Applicable drawings or standards.

Type	Class	MS part number or Army drawing number
I	1	MS51301-1
I	2	MS51301-2
II	1	MS51074-1
II	2	MS51074-2
III	1	8675930
III	2	8675930
IV	-	7728527

3.4 Performance.

3.4.1 Sound performance.

3.4.1.1 Type I. When supplied with the nominal voltage and air pressure inputs specified in table II, horns shall meet the sound intensity, current, and airflow requirements of table II (see 4.7.3.1).

3.4.1.2 Types II, III, and IV. When supplied with the test voltage input specified in table II, horns and buzzers shall meet the sound intensity and current requirements of table II (see 4.7.3.1).

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TABLE II. Sound performance requirements.

Type	Class	Nominal voltage (V dc)	Test voltage (V dc)	Air pressure (kPa)	Minimum sound intensity level [db(A)]	Maximum direct current (A) 1/	Maximum airflow (m^3/s) 2/
I	1 2	24 12	28.8 14.4	620 620	125 125	3 6	0.004 0.004
II	1 2	24 12	28.8 14.4		109 109	3 6	
III	1 2	24 12	28.8 14.4		90 90	3 6	
IV	-	12	14.4		109	6	

1/ Amperes (A).
2/ Cubic meters per second (m^3/s).

3.4.2 Nominal voltage. Horns and buzzers, when operated at the nominal voltage specified in table II, shall produce sound within 5 decibels (A Weighted System Response) [db(A)] of the intensity level specified in table II. Type III, class 1 and 2 volts shall operate at 18 and 9 volts, respectively (see 4.7.3.2).

3.4.3 Reduced air pressure (type I only). Horns shall produce sound within 5 db(A) of the intensity level specified in table II while being operated at reduced air pressures down to 415 kilopascals (kPa) (see 4.7.3.3).

3.4.4 Sound quality.

3.4.4.1 Types I, II and IV. Horns shall have a fundamental frequency between 100 and 500 cycles per second (cps) and the major noise-producing harmonic shall be not greater than 3500 cps when tested in accordance with 4.7.3.4.

3.4.4.2 Type III. Buzzers shall have a fundamental frequency between 105 and 185 cps when tested in accordance with 4.7.3.4.

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3.4.5 Operating cycle.

3.4.5.1 Continuous operation. Horns and buzzers, after 15 minutes of continuous operation, shall meet the requirements of 3.4.1 specified below and the requirements of 3.4.4 (see 4.7.3.5.1):

a. Horns shall meet the requirements of 3.4.1 within 5 db(A) of the intensity level specified in table II.

b. Buzzers shall evidence no deviation from the intensity level specified in table II.

3.4.5.2 Intermittent operation. Horns and buzzers, after being operated for 100,000 cycles, shall meet the requirements of 3.4.1 as specified in 3.4.5.1a, 3.4.5.1b and the requirements of 3.4.4. One cycle of operation shall consist of 1 second "on" (energized) and 2 seconds "off" (see 4.7.3.5.2).

3.4.6 Insulation resistance. The electrical insulation of the horns and buzzers shall evidence no cracking, charring, loosening, burning, smoking or other damage after being subjected to 500 V root mean square (rms) at a frequency of 60 cps applied for 1 minute between each terminal and ground, with the internal grounds disconnected (see 4.7.3.6).

3.4.7 Connector and terminal strength. Horns and buzzers shall meet the requirements of 3.4.8.3 after the connectors and terminals are subjected to a 220-Newton (N) pull (see 4.7.3.7).

3.4.8 Environmental.

3.4.8.1 High temperature. After being stabilized at 71 degrees Celsius (°C), horns and buzzers shall produce sound within 10 db(A) of the intensity level specified in table II (see 4.7.3.8.1).

3.4.8.2 Low temperature. After being stabilized at minus (-) 54°C, horns and buzzers shall produce sound within 10 db(A) of the intensity level specified in table II (see 4.7.3.8.2).

3.4.8.3 Waterproofness (types I, II and III only). Horns and buzzers shall evidence no air bubbles escaping from any sealed chamber when submerged in a salt water solution and shall subsequently meet the requirements of 3.4.1 and 3.4.4 (see 4.7.3.8.3).

3.4.8.3.1 Valve airtightness (type I only). Horns shall evidence no bubbles escaping past the air valve from the air valve inclosure after being connected to a 620 kPa air source and submerged in water (see 4.7.3.8.3.1).

3.4.8.4 Salt fog. Horns and buzzers shall meet the requirements of 3.4.1 and 3.4.4 and shall evidence no corrosion after exposure to a salt fog atmosphere (see 4.7.3.8.4).

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3.4.8.5 Fungus. Horns and buzzers shall show no evidence of fungus growth affecting operation and shall meet the requirements of 3.4.1 and 3.4.4 after exposure to conditions favorable to fungus growth (see 4.7.3.8.5).

3.4.8.6 Vibration. Horns and buzzers shall show no evidence of disintegration of component materials or assemblies and shall meet the requirements of 3.4.1 and 3.4.4 after being subjected to vibration in each direction of the three major axes. The amplitude and frequency shall be as specified in MIL-STD-202, method 201A (see 4.7.3.8.6).

3.4.8.7 Shock. Horns and buzzers shall meet the requirements of 3.4.1 and 3.4.4 after being subjected to induced shocks of sawtooth waveform along each of the three mutually perpendicular axes with a peak value of 100 gravity units (g) (types I, II and III) and 50 g (type IV) for 6 milliseconds (ms) time duration (see 4.7.3.8.7).

3.5 Finish. Unless otherwise specified (see 6.2), the outer surface of the horn or buzzer shall be cleaned in accordance with the applicable method of TT-C-490. The cleaned surfaces shall then be treated with a coating conforming to type I or III of TT-C-490 after which the surfaces shall be primed with primer conforming to MIL-P-53030 or MIL-P-53022, to dry thickness of 0.002 to 0.038 millimeters (mm). The outer surface shall then be painted with a chemical agent resistant coating conforming to color green 383, chip number 34094, per MIL-C-46168 or MIL-C-53039, to a dry thickness ot at least 0.046 mm. All color numbers refer to FED-STD-595 (see 4.7.2).

3.6 Identification and markings. The following information shall be permanently and legibly marked on each horn and buzzer, as applicable. in accordance with MIL-STD-130 (see 4.7.2):

Horn, Air-Operated (Automotive) or, Horn, Electrical (Automotive), or Buzzer, Electrical.
Pressure rating, when applicable.
Military or MS part number.
Manufacturer's part or serial number.
Manufacturer's name or trademark.
Other information deemed necessary by the manufacturer or required on the applicable drawing.

3.7 Workmanship. Workmanship shall be of such quality as to assure that horns and buzzers furnished under this specification are free of defects that compromise, limit, or reduce performance or intended use. Horns and buzzers shall be free of burrs, scratches, rust, chips, projections, and sharp edges or any condition that might present a safety hazard to operating or maintenance personnel (see 4.7.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections:

a. First article inspection (see 4.4).
b. Quality conformance inspections (QCI) (see 4.5).
    1. Examination (see 4.5.1).
    2. Tests (see 4.5.2).
c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 23 ± 10°C
b. Barometric pressure 725 + 50 mm mercury
    - 75
c. Relative humidity 50 ± 30%

4.4 First article inspection. Unless otherwise specified (see 6.2), the Government shall select four horns or buzzers, of the specific type and class (when applicable) produced under the production contract for first article inspection. First article samples shall be inspected as specified in table III, in the order listed in table IV. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply horns and buzzers that are fully representative of those inspected as a first article sample. Any changes or deviations of the production units from the first article sample shall be subject to the approval of the contracting officer.

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TABLE III. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance		
				Examination	Test	Control
Materials and construction	3.2 and 3.3	4.7.1	X			
Defects (see table V)	3.3, 3.5, 3.6 and 3.7	4.7.2	X	X		
Sound performance	3.4.1	4.7.3.1	X		X	
Reduced voltage	3.4.2	4.7.3.2	X			
Reduced air pressure 1/	3.4.3	4.7.3.3	X			X
Sound quality	3.4.4	4.7.3.4	X		X	
Operating cycle	3.4.5					
Continuous operation	3.4.5.1	4.7.3.5.1	X			X
Intermittent operation	3.4.5.2	4.7.3.5.2	X			
Insulation resistance	3.4.6	4.7.3.6	X			X
Connector and terminal strength	3.4.7	4.7.3.7	X			
Environmental	3.4.8					
High temperature	3.4.8.1	4.7.3.8.1	X			
Low temperature	3.4.8.2	4.7.3.8.2	X			
Waterproofness 2/	3.4.8.3	4.7.3.8.3	X			
Valve airtightness 1/	3.4.8.3.1	4.7.3.8.3.1	X			
Salt fog	3.4.8.4	4.7.3.8.4	X			
Fungus	3.4.8.5	4.7.3.8.5	X			
Vibration	3.4.8.6	4.7.3.8.6	X			
Shock	3.4.8.7	4.7.3.8.7	X			

1/ Type I only.
2/ Types I, II, and III only.

TABLE IV. First article inspection sequence.

Sample number			
1	2	3	4
4.7.1	4.7.1	4.7.1	4.7.1
4.7.2	4.7.2	4.7.2	4.7.2
4.7.3.5.1	4.7.3.1		4.7.3.1
4.7.3.1	4.7.3.8.1	4.7.3.1	4.7.3.8.1
4.7.3.8.1	4.7.3.8.2	4.7.3.8.1	4.7.3.8.2

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TABLE IV. First article inspection sequence - Continued.

Sample number			
1	2	3	4
4.7.3.8.2	4.7.3.3 1/	4.7.3.8.2	4.7.3.3 1/
4.7.3.3 1/	4.7.3.2	4.7.3.3 1/	4.7.3.2
4.7.3.2	4.7.3.4	4.7.3.2	4.7.3.4
4.7.3.4	4.7.3.8.3.1 1/	4.7.3.4	4.7.3.8.3.1 1/
4.7.3.8.3.1 1/	4.7.3.5.2	4.7.3.8.3.1 1/	4.7.3.8.7
	4.7.3.1	4.7.3.8.6	4.7.3.1
	4.7.3.4	4.7.3.1	4.7.3.4
	4.7.3.8.3.1 1/	4.7.3.4	4.7.3.8.3.1 1/
	4.7.3.7	4.7.3.8.3.1 1/	4.7.3.8.4
	4.7.3.8.3 2/	4.7.3.8.5	4.7.3.1
	4.7.3.1	4.7.3.1	4.7.3.4
	4.7.3.4	4.7.3.4	
	4.7.3.8.3.1 1/		
	4.7.3.6		

1/ Type I only.
2/ Types I, II, and III only.

4.4.1 First article inspection failure. Deficiencies found during, or as a result of, the first article inspection shall be cause for rejection of the first article sample until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article inspection shall be evidence that all items already produced prior to completion of the first article test are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government will not accept products until first article inspection is completed to the satisfaction of the Government.

4.5 QCI. QCI shall include the examination of 4.5.2 and the tests of 4.5.3. Noncompliance with any of the specified requirements in section 3 and 5 shall be cause for rejection of the sample and the inspection lot.

4.5.1 Sampling plan. Unless otherwise specified (see 6.2), the sampling plan specified herein shall be used. See 6.4 for definition of sampling inspection terms.

4.5.1.1 Lot formation. An inspection lot shall consist of all the horns or buzzers of one type, class and part identification number (PIN), from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

4.5.1.2 Sample. The sample for QCI examination and tests shall be randomly selected from the inspection lot in accordance with table V.

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TABLE V. Sampling plan for QCI.

QCI sampling plan			
Inspection lot size	Sample size		
	Examination		Test
	Major	Minor	
2 to 8	*	5	2
9 to 15	13	5	2
16 to 25	13	5	2
26 to 50	13	5	2
51 to 90	13	7	8
91 to 150	13	11	8
151 to 280	20	13	8
281 to 500	29	16	8
501 to 1200	34	19	13
1201 to 3200	42	23	13
3,201 to 10,000	50	29	20
10,001 to 35,000	60	35	20
35,001 to 150,000	74	40	32
150,001 to 500,000	90	40	32
500,001 and over	102	40	50

* Indicates entire lot must be inspected (100% inspection).

4.5.2 Examination. The sample selected in accordance with 4.5.1.2 shall be examined and defects classified as specified in table VI (see 4.7.2). The acceptance number in all cases is zero.

TABLE VI. Classification of defects.

Category	Defect	Method of examination
Critical	None	
Major		
101	Assembly, incomplete (see 3.3).	Visual
102	Dimensions affecting interchangeability, out of tolerance (see 3.3).	SIR 1/
103	Identification marking, improper (see 3.6).	Visual
104	Faulty workmanship affecting performance (see 3.7).	Visual

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TABLE VI. Classification of defects - Continued.

Category	Defect	Method of examination
Minor		
201	Dimensions not affecting interchangeability, out of tolerance (see 3.3).	SIE
202	Finish, improper (see 3.5).	Visual and functional
203	Faulty workmanship affecting appearance. (see 3.7).	Visual

1/ SIE - Standard Inspection Equipment.

4.5.3 Tests. The sample selected in accordance with 4.5.1.2 shall be subjected to the tests specified in table III. The acceptance number in all cases is zero.

4.5.4 QCI failure. Any item that fails to conform to any specified requirement shall be rejected; any failure (one or more) of the selected sample in either the Major/Minor categories or test for the appropriate inspection lot size shall constitute a failure of the entire lot. The rejected item(s) may be repaired or corrected ans resubmitted for inspection. If the contractor utilizes sampling inspection as an element of his inspection system, rejected inspection lots may be resubmitted for acceptance if the contractor performs 100 percent inspection on the lot for those characteristics which were defective and resulted in rejection of the lot and removes all defective units or obtains procuring activity approval to resample the lot due to the insignificance of the defects. Resubmitted lots shall be kept separate from new lots and shall be clearly identified as resubmitted lots.

4.6 Control tests. Control tests shall be conducted on two horns or buzzers from each lot of 500 units consecutively produced, except that not more than four horns and buzzers shall be performed in a 30-day period, nor less than two horns or buzzers in a 30-day period. The horns and buzzers shall be subjected to the control tests specified in table III.

4.6.1 Failure. Failure of any horn or buzzer to pass any of the specified control tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

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4.7 Methods of inspection.

4.7.1 Materials and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7.2 Defects. Conformance to 3.3, 3.5, 3.6, and 3.7 shall be determined by examination for the defects listed in table VI. Examination shall be visual, functional, or by measurement with standard inspection equipment.

4.7.3 Performance.

4.7.3.1 Sound performance. To determine conformance to 3.4.1, horns and buzzers shall be tested as specified below.

4.7.3.1.1 Apparatus. The test apparatus shall include the following:

a. A power source consisting of an automotive battery of the correct nominal voltage as specified in table II, equipped with an electrical power supply connected in parallel to maintain the state of a constant fully charged battery. The power supply's terminal voltage shall not contain a ripple component of more than 1 V peak to peak, as measured by an oscilloscope, when operating the test specimen.
b. An air source with means for controlling and measuring air pressure and airflow (volume).
c. A sound level meter.
d. A sound analyzer.
e. A voltmeter.
f. An ammeter.
g. A sound level calibrator.

4.7.3.1.2 Procedure.

4.7.3.1.2.1 Type I. Mount the horn so that the sound waves which are emitted from the horn will be directed towards the sound pickup on the sound level meter. The horn shall be not less than 1.2 meters (m) from the floor and 3.0 m from any wall surface so as to minimize sound reflection. The distance between the horn and sound pickup shall be varied to obtain the greatest meter reading possible within the range of 1.5 ± 0.2 m. Ambient temperature shall be maintained at 25 ± 8°C. The horn shall be operated at the test voltage and air pressure specified in table II. Sound intensity, current consumption and airflow shall be measured and recorded to determine conformance to 3.4.1.

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4.7.3.1.2.2 Types II and IV. Horns shall be tested as specified in 4.7.3.1.2.1 while operating at the test voltage specified in table II. Sound intensity and current consumption shall be measured and recorded to determine conformance to 3.4.1.

4.7.3.1.2.3 Type III. Connect the buzzer to the power source. The buzzer shall be suspended freely on a wire with the pickup microphone positioned 10.2 centimeters from the face of, and on the centerline with, the buzzer. The buzzer shall be operated at the test voltage specified in table II. The sound intensity and current consumption shall be measured and recorded to determine conformance to 3.4.1.

4.7.3.2 Nominal voltage. To determine conformance to 3.4.2, horns and buzzers shall be tested as specified in 4.7.3.1 except that the horns and buzzers shall be operated at nominal voltages specified in table II. Type III, Class 1 and 2 shall operate at 18 and 9 volts, respectively. The sound intensity shall be measured and recorded.

4.7.3.3 Reduced air pressure (type I only). To determine conformance to 3.4.3, horns shall be tested as specified in 4.7.3.1 except that the horns shall be operated by air pressures of 415, 485 and 550 kPa. The sound intensity shall be measured and recorded at each air pressure.

4.7.3.4 Sound quality. To determine conformance to 3.4.4, horns and buzzers shall be tested as specified in 4.7.3.1, as applicable, except that the sound level meter shall be connected to a sound analyzer. The frequency of the fundamental and of the most intense harmonic shall be measured and recorded.

4.7.3.5 Operating cycle.

4.7.3.5.1 Continuous operation. To determine conformance to 3.4.5.1, an untested horn or buzzer shall be mounted as in intended use and shall be continuously operated for 15 minutes. At the end of the 15-minute period, the horn or buzzer shall be allowed to return to ambient air temperature and shall be tested as specified in 4.7.3.1 and 4.7.3.4 (see 4.7.3.5.3).

4.7.3.5.2 Intermittent operation. To determine conformance to 3.4.5.2, horns and buzzers shall be mounted as in intended use and operated for 100 000 cycles. At the completion of 100 000 cycles, the horns and buzzers shall be allowed to return to ambient air temperature and shall be tested as specified in 4.7.3.1 and 4.7.3.4 (see 4.7.3.5.3).

4.7.3.5.3 Type I only. Before and after testing as specified in 4.7.3.5.1 and 4.7.3.5.2, hours shall meet the requirements of 3.4.8.3.1.

4.7.3.6 Insulation resistance. To determine conformance to 3.4.6, horns and buzzers shall be subjected to 500 V rms at a frequency of 60 cps, applied 1 minute between each terminal and ground, with the internal grounds disconnected. The horns and buzzers shall then be visually examined.

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4.7.3.7 Connector and terminal strength. To determine conformance to 3.4.7, the connectors and terminals of the horns and buzzers shall be subjected to a 220-N pull for 15 minutes, first in a direction normal to and then in a direction parallel to their respective longitudinal axes. The normal axes pull shall be made from their mating ends. The horns and buzzers shall then be tested as specified in 4.7.3.8.3.

4.7.3.8 Environmental.

4.7.3.8.1 High temperature. To determine conformance to 3.4.8.1, horns and buzzers shall be tested as specified in 4.7.3.1 except that the temperature of the air and the horn or buzzer shall be stabilized at 71°C.

4.7.3.8.2 Low temperature. To determine conformance to 3.4.8.2, horns and buzzers shall be tested as specified in 4.7.3.1 except that the temperature of the air and the horn or buzzer shall be stabilized at -54°C.

4.7.3.8.3 Waterproofness (types I, II and III only). To determine conformance to 3.4.8.3, horns and buzzers shall be tested in accordance with MIL-STD-1184, type II, class 1, method 100, procedure I. After exposure to salt water solution, horns and buzzers shall be tested as specified in 4.7.3.1 and 4.7.3.4.

4.7.3.8.3.1 Valve airtightness (type I only). To determine conformance to 3.4.8.3.1, horns shall be connected to a 620 kPa air source, with the control disconnected. The horns shall be submerged in water for 10 minutes with the projector opening towards the air-water boundary. There shall be no evidence of air bubbles.

4.7.3.8.4 Salt fog. To determine conformance to 3.4.8.4, horns and buzzers shall be tested in accordance with MIL-STD-202, method 101. A 5% salt solution shall be used and the duration of the test shall be 200 hours. After testing, horns and buzzers shall be subjected to the tests specified in 4.7.3.1 and 4.7.3.4.

4.7.3.8.5 Fungus. To determine conformance to 3.4.8.5, horns and buzzers shall be tested in accordance with MIL-F-13927, class 2, method A, except that the performance test shall be conducted at the end of 90 days of incubation only. Horns and buzzers shall be tested for performance as specified in 4.7.3.1 and 4.7.3.4.

4.7.3.8.6 Vibration. To determine conformance to 3.4.8.6, horns and buzzers shall be tested in accordance with MIL-STD-202, method 201A. Motion shall be applied for at least 1 hour in the direction of each of the three major axes. After testing, horns and buzzers shall be subjected to the tests specified in 4.7.3.1 and 4.7.3.4. Type I horns and buzzers shall subsequently be tested as specified in 4.7.3.8.6.1.

4.7.3.8.6.1 Type I only. After testing as specified in 4.7.3.8.6, horns shall meet the requirements of 3.4.8.3.1.

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4.7.3.8.7 Shock. To determine conformance to 3.4.8.7, horns and buzzers shall be mounted as in intended operation and shall be tested in accordance with MIL-STD-202, method 213B, condition I, except the peak value shall be 50 g for type IV. Two induced shocks of sawtooth waveform shall be applied along each of the three mutually perpendicular axes. After testing, horns and buzzers shall be subjected to the tests specified in 4.7.3.1 and 4.7.3.4. Type I horns and buzzers shall subsequently be tested as specified in 4.7.3.8.7.1.

4.7.3.8.7.1 Type I only. After testing as specified in 4.7.3.8.7, horns shall meet the requirements of 3.4.8.3.1.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with MIL-E-17555H.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Horns and buzzers covered by this specification are intended for use as warning signals on tactical vehicles, combat vehicles, or other equipment operating under severe and extreme conditions of shock, vibration, and climate.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Type and class (when applicable) of horn or buzzer (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).
d. If first article inspection is required (see 3.1).
e. Title, number, and date of applicable drawing (see 3.3).
f. If finish should be other than as specified (see 3.5).
g. Responsibility for inspection, if other than as specified (see 4.1).
h. Inspection conditions, if other than as specified (see 4.3).
i. Arrangements for first article inspection and rights of the Government (see 4.4 and 6.3).
j. Sampling plan if other than as specified (see 4.5.1).
k. Control test sample size and frequency for conducting control tests (see 4.6), disposition of lot(s) covered by the control test samples and of defective items.
l. Selection of applicable level and packaging requirements (see 5.1).

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6.3 First article. When first article inspection is required, the contracting officer should provide specific instructions in acquisition documents regarding the category (categories) of first article inspection, sample size for each category and arrangements regarding location for conducting first article inspection, approval of first article inspection results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Definition of terms used in sampling inspection.

a. Classification of defects. A classification of defects is the enumeration of possible defects of the unit of product classified according to their seriousness. A defect is any nonconformance of the unit of product with specified requirements. Defects will normally be grouped into one or more of the following classes: critical, major and minor defects. Also, defects may be grouped into other classes, or into subclasses within these classes.

b. Critical defects. A critical defect is a defect that judgement and experience indicate would result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product, or a defect that judgement and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

c. Critical defective. A critical defective is a unit of product which contains one or more critical defects and may also contain major and/or minor defects.

d. Defective. A defective is a unit of product which contains one or more defects.

e. Formation of lots or batches. The product shall be assembled into identifiable lots, sublots, batches, or in such other manner as may be prescribed (see 1). Each lot or batch shall, as far as is practicable, consist of units of product of a single type, grade, class, size, and composition, manufactured under essentially the same conditions, and at essentially the same time.

f. Lot or batch. The term lot or batch shall mean "inspection lot" or "inspection batch", i.e., a collection of units or product from which a sample is to be drawn and inspected and may differ from a collection of units designated as a lot or batch for other purposes (e.g., production, shipment, etc.).

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g. Lot or batch size. The lot or batch size is the number of units of product in a lot or batch.

h. Major defect. A major defect is a defect, other than critical, that is likely to result in failure, or to reduce materially the usability of the unit of product for its intended purpose.

i. Major defective. A major defective is a unit of product which contains one or more major defects, and may also contain minor defects but contains no critical defect.

j. Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose, or is a departure from established standards having little bearing on the effective use or operation of the unit.

k. Minor defective. A minor defective is a unit of product which contains one or more minor defects but contains no critical or major defect.

l. Presentation of lots or batches. The formation of the lots or batches, lot or batch size, and the manner in which each lot or batch is to be presented and identified by the supplier shall be designated or approved by the responsible authority. As necessary, the supplier shall provide adequate and suitable storage space for each lot or batch, equipment needed for proper identification and presentation, and personnel for all handling of product required for drawing of samples.

m. Representative sampling. When appropriate, the number of units in the sample shall be selected in proportion to the size of sublots or subbatches, or parts of the lot or batch, identified by some rational criterion. When representative sampling is used, the units from each part of the lot or batch shall be selected at random.

n. Sample A sample consists of one or more units of product drawn from a lot or batch, the units of the sample being selected at random without regard to their quality. The number of units or product in the sample is the sample size.

o. Sampling plan. A sampling plan indicates the number of units of product from each lot or batch which are to be inspected (sample size or series of sample sizes) and the criteria for determining the acceptability of the lot or batch (acceptance and rejection numbers).

p. Time of sampling. Samples may be drawn after all the units comprising the lot or batch have been assembled, or samples may be drawn during assembly of the lot or batch.

6.4.2 Recovered materials. "Recovered materials" means materials that have been collected or recovered from solid waste (see 6.4.3).

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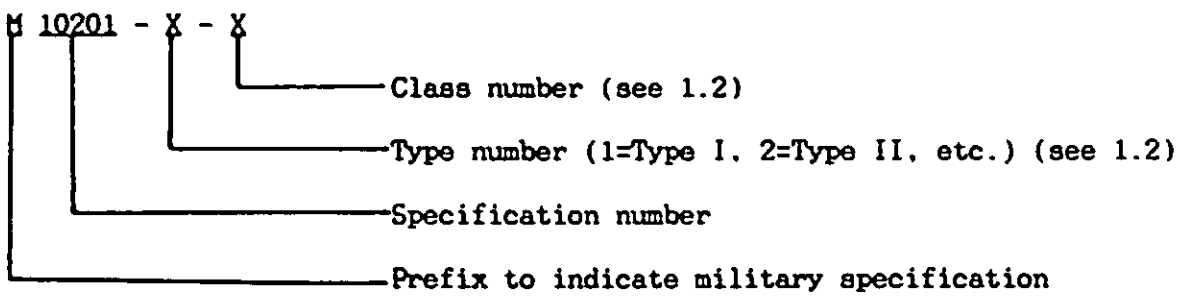
6.4.3 Solid waste. "Solid waste" means (a) any garbage, refuse, or sludge from a waste treatment plant, water supply treatment plant, or air pollution control facility; and (b) other discarded material, including solid, liquid, semisolid, or contained gaseous material resulting from industrial, commercial, mining, and agricultural operations, and from community activities. It does not include solid or dissolved material in domestic sewage, or solid or dissolved material in irrigation return flows or industrial discharges which are point sources subject to permits under section 402 of the Clean Water Act, (33 U.S.C. 1342 et seq.), or source, special nuclear, or byproduct material as defined by the Atomic Energy Act of 1954 (42 U.S.C. 2011 et seq.) (Source: Federal Acquisition Regulations, section 23.402).

6.5 Cross-reference. Table VII provides a cross-reference for classification between this specification and a previous revision (see 1.2).

TABLE VII. Classification cross-reference.

Former classification designation	New classification designation
Type I, class 1, category 1	Type II, class 2
Type I, class 1, category 2	Type II, class 1
Type I, class 2, category 1	Type I, class 2
Type I, class 2, category 2	Type I, class 1
Type II, class 1, category 1	Type III, class 2
Type II, class 1, category 2	Type III, class 1

6.6 Part or Identifying Number (PIN). Except when part numbers are specified on the drawing, MS standard or military specification sheet (MS sheet), the PIN to be used for items acquired to this specification should be developed as follows:



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6.7 Subject term (key word) listing.

Alert mechanism; air, electric, waterproof and non-waterproof
Electrical component
Signal warning, vehicular
Sound warning, vehicular
Vehicle accessory
Warning instrumentation, tactical and combat

6.8 AMC policy on AQLs/LTPDs. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQLs/LTPDs (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Air Force - 99

Review activities:
Army - ER
DLA - CS
Air Force - 84

User activities:
Army - ME
Navy - MC

Preparing activity:
Army - AT

(Project 2590-0219)

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